liquid, and change optical properties when exposed to electrical current. In LCDs, the internal structure involves a layer of liquid crystals being positioned between two polarisers, which an average reader can assume to be a device that allows light to pass at certain angles. Most of the common LCDs work when the crystals change their orientation from twisted to straightened (this particular kind of liquid crystal display, in which the crystals are twisted in their unperturbed orientation, is called a twisted nematic display), which changes the opacity of the crystals to transparent.

If we apply an electric charge to the liquid crystal molecules, they untwist. When they straighten out, they change the angle of the light passing through them so that it no longer matches the angle of the top polarizing filter. Consequently, no light can pass through that area of the LCD, which makes that area darker than the surrounding areas. An LCD that can show colors must have three subpixels with red, green and blue color filters to create each color pixel.

High-resolution color displays, such as modern LCD computer monitors and televisions, use an active-matrix structure. A matrix of thin-film transistors (TFTs) is added to the electrodes in contact with the LC layer. Each pixel has its own dedicated transistor. Active-matrix addressed displays look brighter and sharper than passive-matrix addressed displays of the same size, and generally have quicker response times, producing much better images.

Since these crystals have no internal light of their own, unlike the phosphors in plasma screens, a white fluorescent backlight is used for coloured LCDs, which is positioned behind the screen. When someone refers to an LCD TV, they usually mean a CCFL (cold-cathode fluorescent lamp) backlit LCD screen.

Through the careful control and variation of the voltage applied, the intensity of each subpixel can range over 256 shades. Combining the subpixels produces a possible palette of 16.8 million colors. These color displays take an enormous number of transistors.

But since transistors are known to be fickle and easily damageable devices, one major problem with LCD screens is dead pixels because of defective transistors. This, coupled with problems involving viewing angles, glare levels and contrast ratio, outweigh its numerous advantages related to brightness, power consumption and sharpness of the projected image. Even so, a large number of computer laptop screens, display monitors and smaller consoles use this tech today.